HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Walked industrial sites with the CEO, interviewing utility managers and gathering **mass and energy balance** data for pharmaceutical and medical device facilities.
- Built site mass and energy balances, modeling **heat recovery**, **heat pumps**, and **thermal storage** solutions for live manufacturing sites.
- Authored **Basis of Design studies** and **Measurement and Verification (MV) plans** for energy efficiency projects, directly informing client investment decisions.
- Developed and presented **technical content for proposals** and client calls, contributing to the successful acquisition of new decarbonization projects.
- Presented **energy assessment findings** to site leadership and client boards, initially alongside the CEO and subsequently independently.
- Documented and refined the **FAST assessment methodology**, preparing to teach the process to a newly formed delivery team.
- Managed **production support** for enterprise Java applications on AWS, including **incident response** and SLA adherence using **Splunk** and **Dynatrace**.

SKILLS

Engineering Domain – Industrial Energy, Decarbonization, Mass Balance, Energy Balance, Heat Recovery, Heat Pumps, Thermal Storage, Electrification, Fuel Switching, Measurement and Verification (MV), Basis of Design, Process Heat Integration, Steam Systems, Refrigeration Systems, Compressed Air Systems, HVAC Systems

Programming & Scripting – Python, Java, SQL

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux

Software Development – Spring Boot, REST APIs, Git, Jenkins

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, Pandas, NumPy

Methodologies – Agile, Scrum, Change Management

PROJECTS

Industrial Energy Assessment Framework

- Developed a standardized Python framework for collecting and analyzing site-specific energy consumption data, enabling rapid calculation of **mass and energy balances** for diverse industrial clients.
- Integrated **heat pump and thermal storage** modeling capabilities into the framework, allowing for scenario analysis of electrification and heat recovery options.
- Authored reusable report templates for **Basis of Design studies** and **Measurement and Verification (MV) plans**, streamlining deliverable generation for client proposals.

Energy Systems Monitoring Dashboard

- Constructed an interactive **Power BI dashboard** to visualize real-time energy consumption data from various plant utilities, including steam, compressed air, and refrigeration.
- Implemented **DAX measures** to track key performance indicators such as energy intensity and operational efficiency, providing actionable insights to facility managers.
- Connected to multiple data sources using **Power Query** to aggregate and transform data for comprehensive site energy reporting.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)